Sahaj Bhadja

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Summary

Engineering software developer with 2+ years building simulation, algorithmic, and data-intensive applications across Python, C++, Java, and Linux environments. Experienced in requirements analysis, integration testing, reliability, deployment, and technical documentation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected multi-tenant engineering services on Azure SQL and Blob Storage for 500+ users, applying normalized schemas, validation, and indexed queries to reduce analytical dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant asynchronous workflows for 1,000+ monthly jobs with explicit state transitions, retries, failure isolation, and operational monitoring, improving throughput by 3x while preventing cascading processing failures.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered Python services and REST interfaces handling 10,000+ daily requests, integrating secure ingestion, AWS RDS persistence, and modular processing workflows while maintaining stable performance during peak usage.
- Implemented concurrent processing for 1,000+ uploaded files with validation, error isolation, structured logging, code reviews, and automated tests, reducing feedback turnaround by 60% for 150+ users.

USC AD Labs **Los Angeles, CA**
Machine Learning Researcher *January 2025 – May 2025*
- Built a Decision Transformer and end-to-end simulation pipeline processing 100,000+ trajectory logs, improving long-horizon autonomous-driving planning by 18% through offline reinforcement learning and reproducible closed-loop evaluation.
- Architected a spatial-temporal encoder combining HD-map geometry with ego-agent trajectories through multi-head attention, improving closed-loop simulation stability by 15% while reducing off-road behavior during system-level evaluation.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Engineered cross-platform microservices with object-oriented design, Jenkins pipelines, automated VM builds, and 20+ PyTest and Mockito suites across Windows, Linux, and macOS, strengthening verification and application reliability.
- Optimized API and data-processing workflows using AWS Glue, Amazon S3, SQL, indexing, caching, and pagination, improving application response times by 30% and production pipeline efficiency by 68%.

Pandit Deendayal Energy University **India**
Research Fellow *January 2023 – May 2024*
- Designed a quantum-resistant distributed architecture using five lattice-based cryptographic primitives, analyzing consensus algorithms and benchmark behavior to reduce centralized-control and quantum-risk exposure by 25% in comparative simulations.
- Authored a peer-reviewed technical paper and presented findings at IISc Bengaluru, translating algorithm design, experimental methodology, validation results, and engineering tradeoffs into clear recommendations for technical audiences.

Skills

Languages & Systems: Python, C++, Java, JavaScript, SQL, Bash, Linux, UNIX, object-oriented programming
Modeling & Analysis: simulation, algorithm development, reinforcement learning, data pipelines, statistical analysis, performance evaluation
Testing & Assurance: unit testing, integration testing, PyTest, Mockito, static analysis, requirements analysis, verification, reliability engineering
Tools & Platforms: Git, GitHub, Jira, Jenkins, Docker, CI/CD, IDEs, compilers, AWS, Azure, PostgreSQL, SQL Server
Engineering Practices: systems engineering, system design, debugging, technical documentation, Agile, code reviews, deployment, operational support, risk analysis

Projects

CPU Scheduling Simulator | *C++, React, JavaScript, EJ-2*
- Implemented six scheduling algorithms with deterministic simulation, interactive parameter controls, edge-case testing, and Gantt visualization across 1,000+ process combinations, reducing debugging time by 20% while validating timing metrics.

Linguistix | *Python, Linux, AWS, Docker, CodePipeline*
- Developed a containerized data-processing application with automated releases, validation checks, structured error handling, and scalable REST interfaces, achieving 87% accuracy while strengthening reproducible deployment and production reliability.

CaptionCraft | *Python, PyTorch, Transformers, Flask*
- Benchmarked BLIP, GIT, and ViT-GPT-2 architectures across 10,000 images and 100,000 captions, improving caption accuracy from 81% to 96% while building tested inference interfaces for real-time evaluation.